# Axial Seamount bottom pressure, reservoir loading, and conditional crustal failure in PyLith

Axial Modeling Project

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**Abstract**

Seafloor uplift at Axial Seamount records reservoir deformation, whereas the pressure and stress that drive rupture remain hidden. We use processed Center bottom-pressure-recorder (BPR) observations and a three-dimensional PyLith model to relate the 1998–2011, 2011–2015, and post-2015 uplift histories to reservoir loading. The geometry, thermal and viscosity laws, and joint tensile–shear failure test follow Cabaniss et al. (2020) where their inputs are specified. Five cases use elastic, uniform Maxwell, temperature-dependent Maxwell, hydrothermal Maxwell, and tectonic-plus-hydrothermal loading. We fit deployment-relative uplift without published model outputs. On the 150 m mesh, the elastic screen first reaches joint failure on 21 November 2012, 884 days before the observed 2015 eruption. Its 1998–2011 crossing lies inside an unobserved BPR gap. The historical uniform Maxwell fit also crosses only in that gap, first under negative fitted pressure. The three Maxwell fits for 2011–2015 have no joint crossing through their final 22 April 2015 BPR state. Post-2015 BPR loading crosses the elastic screen on 24 January 2017. These are conditional stress screens because the BPR datums, basal boundary, Maxwell spectrum, and mechanical time step remain unresolved.

## 1 Physical question and observations

Reservoir inflation deforms the seafloor and changes stress around a magma chamber. A BPR depth series constrains vertical motion; it does not directly measure magma pressure or rock failure. Cabaniss et al. [1] connected the Axial record to these hidden quantities with finite elements. We reconstruct that sequence in PyLith to ask whether observed uplift loads a cavity and connected crustal shear path to failure.

The input observations are processed Center records in the archives cited by Cabaniss et al. [3, 4]. We select actual hourly values at a seven-day grid from 13 April 2011 to 22 April 2015, before the 24 April eruption. The 1998–2011 reconstruction uses the same archived Center deployments on a seven-day grid and has a 708-day observational gap after 1998. The post-2015 Maxwell fits also use seven-day steps. Each deployment has an unresolved vertical datum. For pressure inference, its first selected depth defines zero uplift separately. The historical fit also uses a stated 1998–2000 gap uplift to tie the pre-eruption levels. The 0.767 m gap-edge assumption corresponds to 0.755 m between the sampled weekly grid endpoints after their observed edge changes are included. Inferred pressure for 2011–2015 is relative to 13 April 2011 rather than absolute reservoir pressure.

For the current monitoring interval we use the Ocean Observatories Initiative (OOI) Central Caldera MJ03F daily mean seafloor-depth product [6]. The product is detided and drift-corrected by OOI. Its deepest observed day within 25 April–8 May 2015, 4 May, defines zero relative uplift. The public record used here contains 3,728 daily values through 17 October 2026 and 457 missing days; all returned daily-depth quality flags are 2, meaning "not evaluated." No observation is filled across a gap.

Figures 1–5 follow the topics and order of Figures 1–5 in Cabaniss et al., but show our available observations and model states. The first two figures establish the observed setting and BPR input; Figures 3–5 show our material fields and mechanical calculations.

Figure 1 locates the Center BPR in an independently sourced map and relates the summit bathymetry to the actual PyLith mesh. The model origin is placed at the archived 2011–2013 Center coordinate for display; its absolute geographic placement is not determined by the mesh. The low seismic-velocity outlines are project-derived proxies, not the published magma-reservoir boundaries.

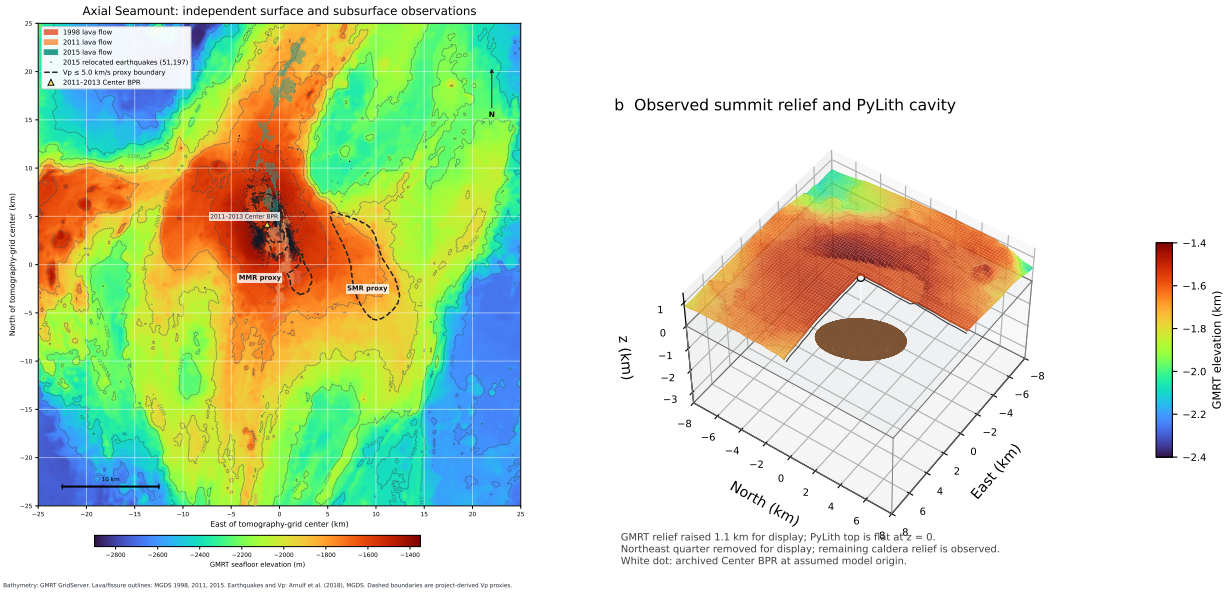



Figure 1: Axial setting and PyLith geometry, following Cabaniss et al. Figure 1. (a) Bathymetry comes from GMRT; lava-flow and fissure outlines, 2015 relocated earthquakes, and the P-wave velocity grid come from cited MGDS records [5]. The yellow marker is the archived 2011–2013 Center BPR. Dashed low-velocity boundaries are project-derived from a smoothed 3.5 km below-sea-level velocity slice at 5.0 km s$^{-1}$; they are not official magma-reservoir outlines. (b) The same GMRT grid lies above the meshed cavity surface from the 33,980-tetrahedron, 150 m PyLith model. The panel shows a 16 by 16 by 3.5 km summit crop of the full 50 by 50 by 10 km domain. The model origin is placed at the Center BPR for display only. Observed relief is raised 1.1 km. The northeast quarter is removed to expose the cavity while preserving the other three quarters of the caldera relief. The vertical display scale is exaggerated about 2.2-fold; the model's solved seafloor boundary remains flat at $z = 0$. The model cavity has no independently established geographic position. No Cabaniss map pixels or modeled fields were used.

Figure 2 stitches the archived Center records for display. The 1998, 2011, and 2015 deflation events occur within recorded deployments; the processing does not impose their drops. Two overlaps constrain relative shifts with paired noon observations. Other deployment offsets assume no net deformation across short unobserved gaps. Across the 708-day 1998–2000 gap, we impose an equal pre-1998 and pre-2011 level scenario. It requires 0.767 m of gap uplift given our archived noon samples. That amount is an assumption, since the relative 1998–2011 level is unobserved [2]. The paper's Figure 2 also includes earthquake counts; a compatible complete daily series is not available here.

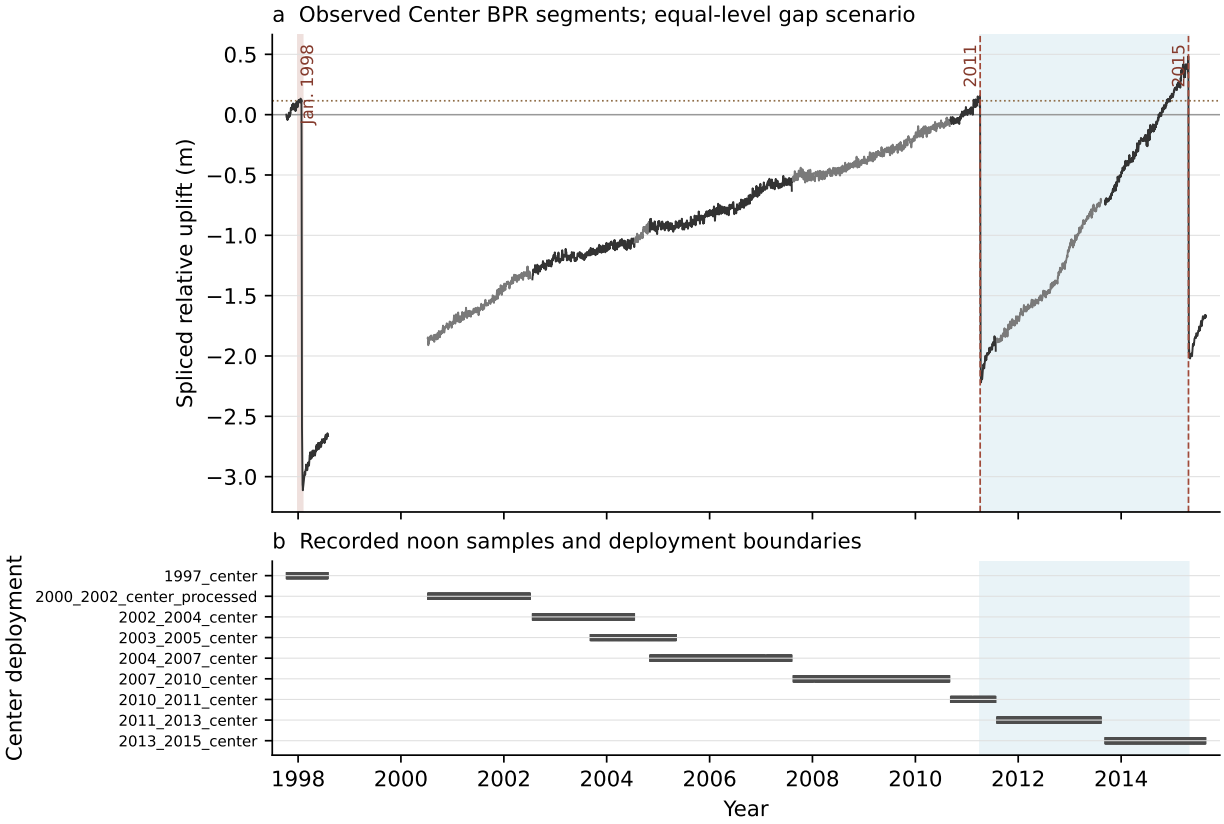



Figure 2: Processed 1997–2015 Center BPR observations and coverage, following the deformation role of Cabaniss et al. Figure 2. Panel (a) uses recorded noon depths from the cited Fox and Chadwick–Nooner archives [4, 3]. Two noon values are removed from each deployment end to reduce recorder-boundary transients. Overlapping deployments are shifted by the median difference of 30 paired noons near each handoff. Nonoverlapping records are aligned by seven-noon medians at gap edges. Short gaps assume zero unobserved deformation; the 1998–2000 gap instead receives 0.767 m under the equal pre-eruption level scenario. The dotted horizontal line shows that assumed common level. Lines stop across gaps, and no observations are interpolated. Panel (b) marks retained noon samples and deployment coverage. The January 1998 shaded month and the 2011 and 2015 lines mark observed eruptions; blue marks the 2011–2015 modeled cycle. The 1997–1998 Fox channel lacks drift correction. The historical model fits use the stated gap uplift as a separate constraint, not as an invented BPR sample. No earthquake-count series is inferred.

## 2 Model and inference

The cavity is a 6 by 3 by 1 km ellipsoid centered 1.6 km below the seafloor and oriented N30°W, following Cabaniss et al. [1]. The 50 by 50 by 10 km domain and fixed base are implementation choices; the source specifies a Winkler foundation but omits domain dimensions. Lateral boundaries have zero normal motion. The 150 m near-cavity mesh has 33,980 tetrahedra and a 0.66% cavity-volume deficit relative to the ideal ellipsoid. It supplies Figures 1b and 3–5 and the post-2015 stress history. PyLith pressure increments are added to gravity-equilibrated reference stress. Some continuation runs use 16 MPI ranks while the gravity reference uses eight. Exact vertex coordinates and tetrahedral vertex sets align their saved stress fields without spatial interpolation. Four cases omit cycle-long tectonic loading. A separate case imposes 60 mm yr$^{-1}$ full N60°E extension across the 68.3 km projected box width, then removes its modeled uplift before fitting pressure. The azimuth is our approximation of ridge-normal spreading. The paper's main text states the full rate, whereas its Table S1 lists boundary velocities of ±20 mm yr$^{-1}$. Initial viscous strain is zero.

Figure 3 of Cabaniss et al. labels the uniform Maxwell viscosity $10^{16}$ Pa s. Its uniform modulus color suggests 60 GPa, which Table S1 gives for an analytical solution; the FEM value is not separately tabulated. The elastic

reference uses the project's existing 50 GPa model. Thermal cases use the printed supplementary laws

$$\eta(T) = A_D \exp\left(\frac{E_A}{R_g T}\right), \qquad E(T) = E_D + \frac{E_B}{1 + C_S \exp\{A_S(1 - T/T_{\max})\}}. \tag{1}$$

Viscosity uses kelvin, $A_D = 10^9$ Pa s, $E_A = 1.2 \times 10^5$ J mol$^{-1}$, and $R_g = 8.3114$ J mol$^{-1}$ K$^{-1}$. For modulus, $E_D = 25$ GPa, $E_B = 50$ GPa, $A_S = 12$, $C_S = 5$, and $T_{\max} = 1200$°C. The cavity is at 1200°C, external geotherm at 30°C km$^{-1}$, internal heat production at zero, and $k_0 = 3$ W m$^{-1}$ K$^{-1}$. The hydrothermal conductivity law uses Nu= 8, $A = 0.75$, $T_{\max} = 600$°C, and $z_{\max} = 6$ km. The printed modulus law increases with temperature, contrary to the source's brittle/ductile labels and property-map colors. We retain the literal law.

PyLith uses one Maxwell branch at these absolute viscosities. The source describes a generalized Maxwell body but does not publish branch weights or relaxation times. A unit pressure ramp gives each case's surface response; we fit pressure changes to deployment-relative BPR motion and run the inferred history forward. The fit penalizes squared second differences of weekly pressure at weight 1; weights 0.1 and 10 test gap sensitivity. These are reconstruction choices. No published pressure, volume, stress, or failure output enters the fit.

Joint failure requires positive maximum principal stress at the reservoir for zero tensile strength and a face-connected Mohr–Coulomb shear-yield path from cavity to seafloor [1]. The shear screen uses source values of 1 MPa cohesion and 25° friction angle. Zero pore pressure is a reconstruction assumption. Separate elastic sensitivity scans use 0.1 and 2.5 MPa tensile strength; Figure 5 uses zero. Following the paper's criterion, one tensile element on the cavity boundary suffices. In its terminology, cavity tension alone makes a model "eruptible"; the reported eruption-screen dates require the connected shear path as well. The paper does not prescribe a discrete connectivity rule; shared tetrahedral faces define a path in this reconstruction.

## 3 Calculated properties and loading

Figure 3 uses the layout of Cabaniss et al. Figure 3 with our calculated material fields. The literal modulus law gives 25.0–33.3 GPa on the 150 m mesh, and the minimum cell viscosity is $1.80 \times 10^{13}$ Pa s. Temperature contours and conductivity are calculated from the local thermal solves, not traced from the paper.

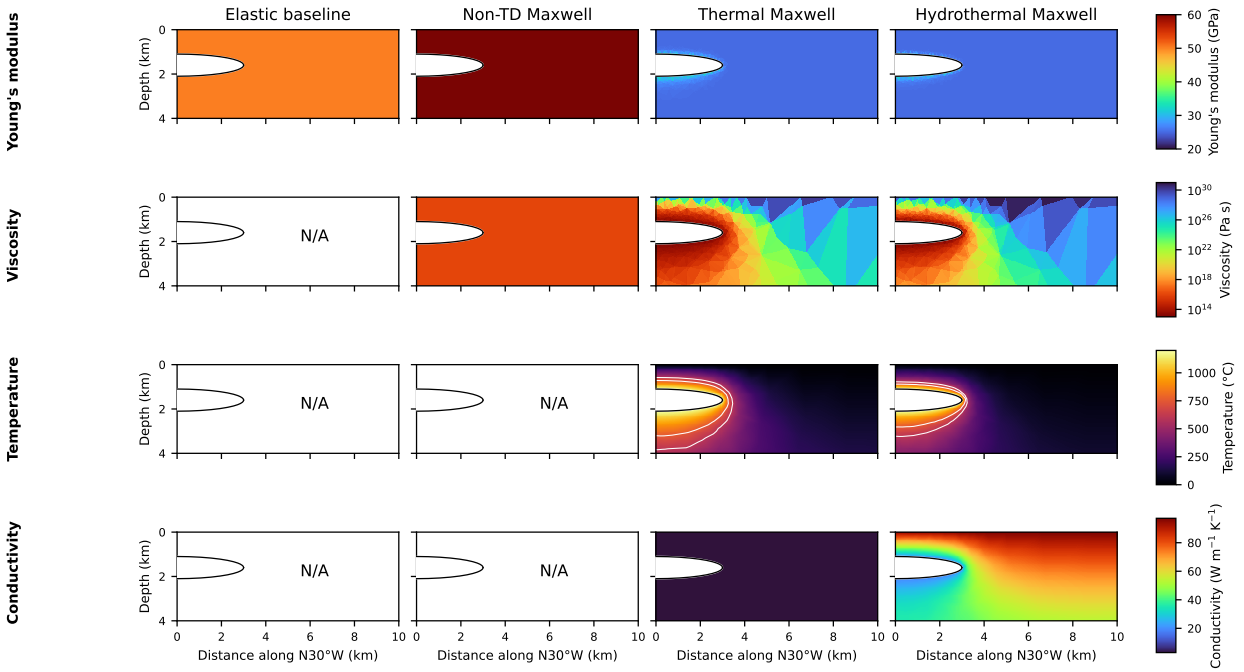



Figure 3: PyLith property sections in the four-column, four-row layout of Cabaniss et al. Figure 3. Columns are the existing elastic reference, uniform Maxwell, thermal Maxwell, and hydrothermal Maxwell cases. Rows are modulus, viscosity, temperature, and conductivity. Source inputs and local thermal fields use the 150 m PyLith mesh. Temperature is interpolated from saved thermal nodes inside tetrahedra; modulus and viscosity are the assigned cellwise values along the N30°W major axis. Row labels appear at left. White contours mark 600 and 750°C; the cavity outline is the stated ideal geometry. N/A denotes an unused property. The elastic reference is 50 GPa; the 60 GPa uniform Maxwell modulus is an explicit figure-reading choice. No published output fields were used.

Figure 4 places observed uplift beside inferred pressure and calculated cavity expansion, following Cabaniss et al. Figure 4. Each rheology fits the same deployment-relative observations. Volume is integrated from displacement on the meshed cavity at every saved time. The colored pressure bands vary only the curvature penalty; they are not posterior confidence intervals. The sharp modeled decline near the 2013 deployment change is a consequence of fitting separate, unknown observation datums across a gap. It is not a continuous observed deflation curve. Consequently, the pressure and volume change across that interval depend on the fit's smoothing assumption.

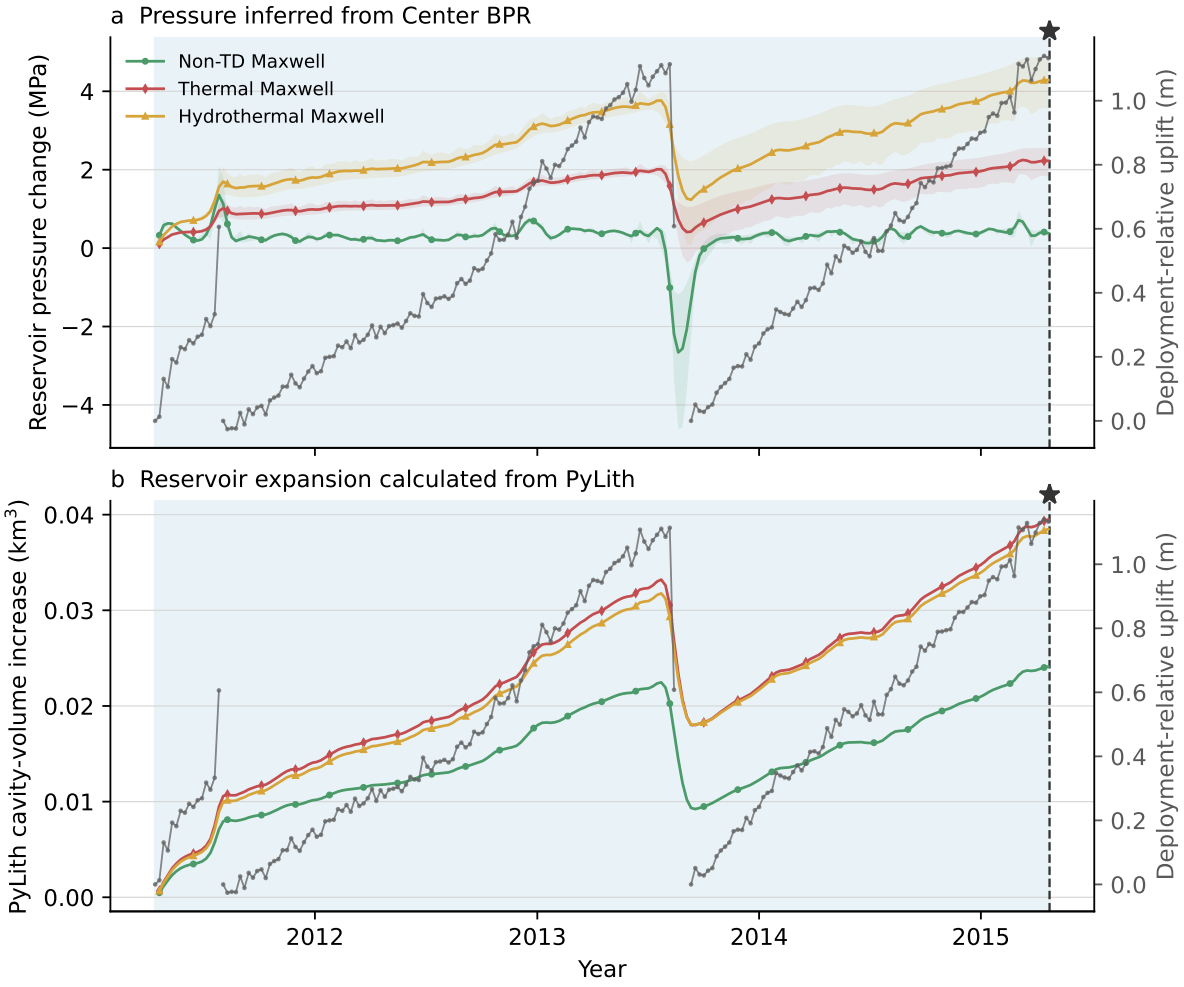



Figure 4: Calculated 2011–2015 reservoir pressure (a) and cavity-volume increase (b), with processed Center BPR uplift. Green circles, red diamonds, and gold triangles denote uniform, thermal, and hydrothermal Maxwell cases. Gray observations are plotted separately for each deployment relative to its first selected depth; their offsets are unknown. The right axis applies only to gray data. Color bands in (a) span curvature weights 0.1–10; solid curves use weight 1. Volume is a first-order integral of PyLith cavity displacement. The star and dashed line mark the observed 24 April 2015 eruption, not a predicted failure. Blue shading marks the modeled interval. Published output curves were neither digitized nor fitted.

## 4 Failure and numerical limits

The elastic 2011–2015 history first meets the cavity-tension and connected shear-path screens on 21 November 2012 at 12.15 MPa, 884 days before the observed eruption. A rotated vertical section intersects 16 cells on the saved three-dimensional path from reservoir to seafloor. The first tensile cell centroid is 1.52 km from the reservoir's major axis at 1.54 km depth, on the meshed cavity flank. The 1998–2011 elastic fit first crosses on 13 July 1999 at 12.11 MPa, 4,285 days before the observed 2011 eruption but inside the unobserved 1998–2000 gap. Its timing depends on the assumed uplift across that gap. The three 150 m Maxwell fits for 2011–2015 have neither a connected shear path nor a tensile cavity element through their last BPR sample on 22 April 2015. Their separate two-day continuations to 24 April also have no joint failure. The uniform, thermal, and hydrothermal cavity maximum principal stresses at that observed date are −26.34, −25.85, and −30.16 MPa, respectively. The spreading-plus-hydrothermal refit gives 4.33 MPa at the observed date and −30.13 MPa maximum cavity principal stress, also without joint failure. The historical uniform Maxwell fit first reaches the joint screen on 19 January 1999 at a negative fitted pressure change of −14.78 MPa. Its face-connected path contains 20 model

cells; the rotated section cuts 18. Its first positive-pressure joint state is on 4 January 2000 at 12.56 MPa. Both dates fall in the unobserved 1998–2000 interval and cannot be treated as data-constrained forecasts. At the final pre-2011 fitted state, its cavity maximum principal stress is −17.49 MPa and the joint screen is absent. Holding that fitted pressure to the observed eruption date leaves the joint screen absent, with a cavity maximum principal stress of −17.51 MPa. The full temperature-dependent historical fit has no connected shear path or tensile cavity state in its weekly history. Its final fitted pressure change is 3.31 MPa and its cavity maximum principal stress is −20.47 MPa. Holding that pressure to 6 April 2011 also leaves no joint failure, with a cavity maximum principal stress of −20.47 MPa. The historical hydrothermal fit likewise has no connected shear path or tensile cavity state in its weekly history. Its final pressure change is 6.49 MPa and its cavity maximum principal stress is −25.62 MPa. Holding that pressure to 6 April 2011 leaves no joint failure and a cavity maximum principal stress of −25.63 MPa. The separate 60 $\mathrm{mm\,yr^{-1}}$ spreading sensitivity refits the 1998–2011 pressure change to 6.62 MPa at the eruption date. It has no cavity tension or connected shear path through that date; its cavity maximum principal stress is −25.30 MPa. Figure 5 also shows separately calculated eruption-date states for all five cases, including the tectonic sensitivity.

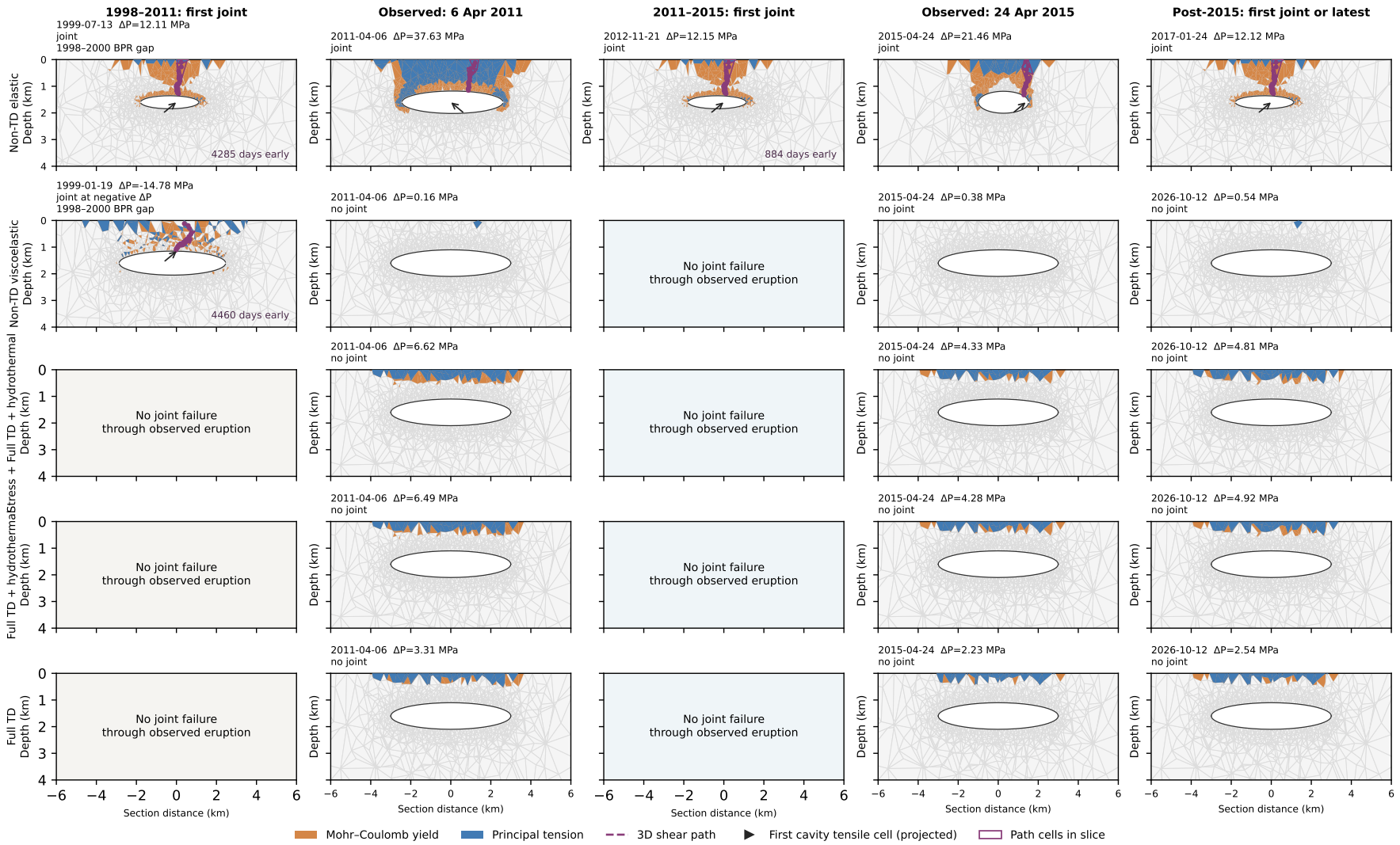



Figure 5: Five PyLith cases in the order of Cabaniss et al. Figure 5. Each historical pair shows the first sampled joint crossing beside the observed eruption-date state; the last column shows the first or latest post-2015 state. Blank panels have no joint crossing. Orange marks Mohr–Coulomb yield and blue marks principal tension. Joint crossing requires one tensile cavity cell and a face-connected shear path to the surface. Black arrows project the first tensile cavity cell; magenta outlines mark path cells cut by the section, and the dashed line projects the three-dimensional path. At joint crossing, the section rotates to intersect the saved path; other states use the central N30°W plane. Off-center sections show a smaller cut through the same cavity. First crossing labels give retrospective days before eruption, not prospective forecasts. Observed-date runs hold the final fitted pressure to the eruption date. All fields come from local PyLith output.

Table 1: Center BPR fits and cavity stress at the last weekly fitted state, 22 April 2015, on the 150 m mesh. Pressure is relative to 13 April 2011; root-mean-square (RMS) error uses recorded BPR samples only.

| Rheology | RMS (m) | Last $\Delta P$ (MPa) | Cavity $\sigma_1$ (MPa) |
|---|---|---|---|
| Uniform Maxwell | 0.02294 | 0.379 | −26.35 |
| Hydrothermal Maxwell | 0.02226 | 4.283 | −30.16 |
| Full TD Maxwell | 0.02223 | 2.226 | −25.85 |

The last modeled first-order cavity-volume increases are 0.02412, 0.03950, and 0.03849 $km^3$ for the uniform, thermal, and hydrothermal cases, respectively.

The hottest cells have a Maxwell time near 1,354 s, compared with the 604,800 s step in the four-year fit, a factor of about 447. The current weekly history therefore does not resolve the shortest viscoelastic relaxation time. Temporal and failure-path convergence remain open.

# 5 Post-2015 observed loading

Figure 6 maps the observed OOI depth changes to the same PyLith cavity and failure screen. Relative uplift reaches 2.672 m on the last available day. The elastic stress screen first meets both joint conditions on 24 January 2017 at an inferred 12.12 MPa pressure change. The elastic stress state has a 16-cell face-connected shear path at that date. The four Maxwell cases, including the separate hydrothermal spreading run, do not meet both conditions through their last weekly state on 12 October 2026. Their maximum cavity principal stresses over that interval remain negative: −23.13, −23.29, −27.94, and −27.93 MPa for uniform, thermal, hydrothermal, and spreading plus hydrothermal Maxwell, respectively. Thus the elastic configuration crosses during continued observed reinflation, whereas the Maxwell configurations give no crossing date within the observed window. These dates are retrospective BPR-constrained failure screens, not forecasts of a future eruption.

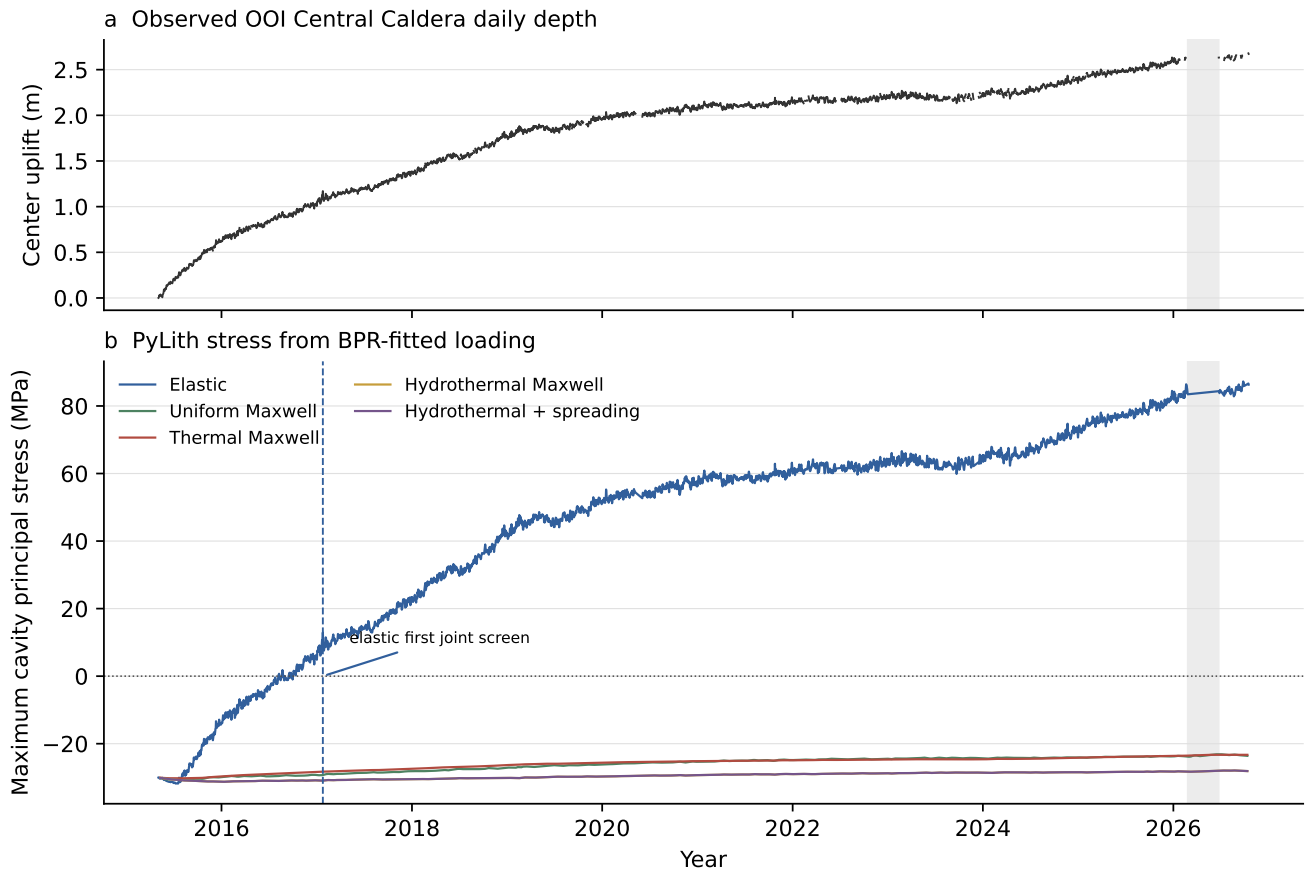



Figure 6: Observed OOI Center uplift and calculated cavity stress since the 2015 eruption. (a) Recorded MJ03F daily depth differences from the deepest post-eruption day, 4 May 2015. Lines stop across missing days; gray bands mark observation gaps of at least 30 days. (b) Maximum principal stress at the PyLith cavity boundary under pressure inferred separately for each rheology from the same Center BPR history. The blue elastic curve uses the saved 1 MPa PyLith stress response; the green, red, gold, and purple curves use forward uniform, thermal, hydrothermal, and spreading plus hydrothermal Maxwell solves. The spreading case refits pressure after removing the modeled tectonic displacement. The dotted horizontal line marks zero tensile strength, a necessary but insufficient part of the joint criterion. The dashed blue line marks the first sampled elastic state with both cavity tension and a connected Mohr–Coulomb path. No Maxwell case satisfies both conditions through 12 October 2026. Their curves span observational gaps through the stated curvature penalty; they are model fits, not missing measurements or prospective forecasts.

# 6 Interpretation

The model maps BPR deformation to pressure, cavity volume, and stress without importing the paper's output data. It does not reproduce the paper's modeled eruption state: the elastic threshold is reached in 2012, and the Maxwell cases remain below joint failure in April 2015. The mismatch could reflect the single Maxwell branch, initial state, thermal modulus conflict, basal treatment, observation datums, or resolution. The published numerical inputs do not isolate these effects. A defensible threshold requires constrained gaps and initial state, a resolved Maxwell spectrum and base response, and converged cavity stress and path connectivity. The 1998–2011 cycle and a ridge-normal tectonic sensitivity are included here; neither resolves the mismatch at both eruption dates. The 150 m calculation is a finer cavity mesh, but stress and path convergence across meshes remain untested. We do not draw a Figure 6-style eruption pathway because these stress screens do not simulate fracture growth or magma transport.

# 7 Reproducible record

The BPR, failure-threshold, and figure targets documented in the repository README regenerate the histories and figures from cited BPR inputs and PyLith 5.0.2. Local summary files record source and output digests. Raw observations, meshes, and PyLith outputs are untracked; the source audit and parameter record state the remaining assumptions. The six figures here use cited observations, model inputs, or generated results; none copies or digitizes a published panel.

# References


[1] H. E. Cabaniss, P. M. Gregg, S. L. Nooner, and W. W. Chadwick, *Triggering of eruptions at Axial Seamount, Juan de Fuca Ridge*, Scientific Reports 10, 10219 (2020). doi:10.1038/s41598-020-67043-0.

[2] W. W. Chadwick et al., *Geodetic monitoring at Axial Seamount since its 2015 eruption reveals a waning magma supply and tightly linked rates of deformation and seismicity*, Geochemistry, Geophysics, Geosystems 23 (2022). doi:10.1029/2021GC010153.

[3] W. W. Chadwick and S. L. Nooner, Axial Seamount seafloor pressure data, doi:10.1594/IEDA/322282.

[4] C. G. Fox, Axial Seamount seafloor pressure data, doi:10.1594/IEDA/322344.

[5] Global Multi-Resolution Topography, GMRT GridServer; Marine Geoscience Data System lava-flow outlines for 1998, 2011, and 2015; 2015 relocated earthquakes; and P-wave velocity.

[6] Ocean Observatories Initiative, Central Caldera MJ03F bottom-pressure and tilt instrument, daily mean seafloor depth (BOTSFLU-DAYDEPTH), accessed 23 October 2026. OOI ERDDAP dataset.